

Double strands without a new lemma: the reverse-complement double cover of a de Bruijn graph

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Abstract

The single-strand theory of the companion notes sees a repeat only between co-oriented copies, so the seven ribosomal operons of *E. coli* — five on one strand, two on the other — appear as a five-copy family and a two-copy family and never as a seven-copy one. Repairing this was expected to need a sandpile theory for bidirected graphs, which does not exist. It does not need one. Let $D_k(S)$ be the multidigraph whose arcs are the $(k+1)$ -mer occurrences in S and in its reverse complement $\bar{S}$. Then $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$ and $d^+(v) = d^-(v) = \text{mult}_D(v)$: D_k is balanced, so the unitig lemma, the bundle-chain lemma and the splitting theorem apply to it verbatim. Three consequences. *One*: multiplicities add across strands, so a family with a forward and b reverse copies has multiplicity $a + b$ in D_k ; for *E. coli* the longest multiplicity-seven chain has span 728 bp and its seven occurrences are the seven operons, one each, the two minus-strand ones on the reverse strand of D_k . *Two*: D_k is connected exactly when S contains an inverted repeat of length $\geq k$, so the inverted repeats are precisely what glues the strands; φ X174 already qualifies at $k = 12$, through ATTAAGCTCATT at 441 and its reverse complement at 1448. *Three*: the involution $\rho(v) = \bar{v}$ reverses arcs, hence carries D_k to its opposite; since $\Delta^{\text{op}} = \Delta^T$ for a balanced digraph and the two Bowen–Franks cokernels are dual, ρ dualises $\mathcal{K}(D_k)$ instead of splitting it — which is what the warning against ± 1 -eigenspace decompositions was pointing at. The phage’s celebrated single global bit is a single-strand phenomenon: $\mathcal{K}(G_{12}) = \mathbb{Z}/2$ but $\mathcal{K}(D_{12}) = \mathbb{Z}/33$. Six checks, all exact. What is *not* done: D_k is the double cover, not the bidirected quotient, and $\mathcal{K}(D_k)$ does not count double-stranded reconstructions; that problem is left exactly where it was.

1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ has the distinct k -mers as vertices and the occurrences of $(k+1)$ -mers as arcs, and $\mathcal{K}(G_k(S)) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk})$ by the bundle-chain splitting theorem [4]. The theory is single-strand throughout [2, 3, 4, 5], and [6] measured what that costs on a real genome: of the seven *rrn* operons of *E. coli* K-12, five lie on the $+$ strand and two on the $-$, and since a repeat of $G_k(S)$ requires co-oriented copies, the operons split into a multiplicity-five family and a multiplicity-two family, with multiplicity seven occurring nowhere.

The obvious repair is to work with the bidirected graph that assemblers actually build, whose vertices are the classes $\{v, \bar{v}\}$; [2, 3] left this open and warned against the tempting shortcut of decomposing $\mathcal{K}$ into ± 1 -eigenspaces of the reverse-complement involution, since

an involution splits an abelian group only after 2 is inverted and the 2-torsion is where the interesting phenomena live. That warning stands. What this note observes is that most of what one wants from the bidirected picture is already available, without any new lemma, from a balanced digraph.

2 The double cover

Definition 1 (Reverse-complement double cover). Let $\bar{S}$ be the reverse complement of S , again circular of length N . Let $D_k = D_k(S)$ be the multidigraph whose vertices are the k -mers occurring in S or in $\bar{S}$ and whose arcs are the occurrences of $(k+1)$ -mers in S together with those in $\bar{S}$; it has $2N$ arcs. Write $\rho(v) = \bar{v}$.

Lemma 2 (Balance). $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$ and $d^+(v) = d^-(v) = \text{mult}_D(v)$ for every vertex. Hence D_k is a balanced multidigraph.

Proof. $\text{mult}_{\bar{S}}(v) = \text{mult}_S(\bar{v})$, because a copy of v in $\bar{S}$ is a copy of $\bar{v}$ in S ; the first formula follows. In each of the two circular sequences separately, every occurrence of v contributes one arc out of v and one arc into v , as in [2, Lem. 4]; summing the two gives the second. $\square$

Corollary 3 (Nothing new is needed). *The unitig contraction lemma [2, Lem. 4], the bundle-chain lemma [4, Lem. 2], the canonicity of the skeleton [4, Lem. 6] and the splitting theorem [4, Thm. 5] apply to D_k verbatim, and a vertex of D_k survives unitig contraction if and only if $\text{mult}_S(v) + \text{mult}_S(\bar{v}) \geq 2$ [6, Lem. 2.1].*

That last clause is the practical point: the branch vertices of D_k are the k -mers that are repeated *when both strands are counted*, so the entire fast pipeline of [6] runs on D_k unchanged, with the positions of $\bar{S}$ appended to those of S .

Proposition 4 (The strands are glued by the inverted repeats). *D_k is connected if and only if some k -mer w occurs in S together with $\bar{w}$; otherwise $D_k = G_k(S) \sqcup G_k(\bar{S})$, two isomorphic components.*

Proof. The arcs coming from S form a closed walk through every k -mer of S , so those vertices lie in one component, and likewise for $\bar{S}$; the two vertex sets are the k -mers of S and their reverse complements. They meet exactly when some w and $\bar{w}$ both occur in S . $\square$

So the double cover is interesting precisely to the extent that the genome has inverted repeats of length at least k . Both of our genomes qualify: ϕ X174 at $k = 12$ through ATTAAGCTCATT at position 441, whose reverse complement sits at 1448; *E. coli* at $k = 150$ through a 150-mer at 19 796 whose reverse complement is at 3 584 046 (check (C)).

3 Multiplicities add across strands

Theorem 5 (Strand merging). *Let W be a sequence occurring a times on the forward strand of S and b times on the reverse, with $a + b \geq 1$ and W not equal to $\bar{W}$. In $G_k(S)$ the k -mers of W have multiplicity a and those of $\bar{W}$ multiplicity b . In D_k both have multiplicity $a + b$, and the two families are exchanged by ρ . If, moreover, the $a + b$ copies agree over a window of $s + k - 1$ letters, that window contributes a bundle chain of multiplicity $a + b$ and length s to D_k , hence a summand $(\mathbb{Z}/(a + b))^{s-1}$ to $\mathcal{K}(D_k)$, and another from its ρ -image.*

Proof. A reverse-strand copy of W is a forward occurrence of $\bar{W}$, so $\text{mult}_S(w) = a$ and $\text{mult}_S(\bar{w}) = b$ for a k -mer w of W in general position; by Lemma 2, $\text{mult}_D(w) = a + b$, and symmetrically for $\bar{w}$. The last sentence is Lemma 2 followed by [4, Lem. 2] applied to the balanced digraph D_k . $\square$

Corollary 6 (The seven operons). *For E. coli K-12 the seven rrn operons carry, in D_k , bundle chains of multiplicity seven. At $k = 150, 100, 51$ the longest such chain has span 728 bp and its seven occurrences are the seven operons, one each:*

$$\begin{aligned} &+ 224\,044 \rightarrow 223\,771, & - 2\,730\,735 \rightarrow 2\,726\,069, & - 3\,428\,340 \rightarrow 3\,423\,423, \\ &+ 3\,942\,081 \rightarrow 3\,941\,808, & + 4\,035\,804 \rightarrow 4\,035\,531, & + 4\,166\,932 \rightarrow 4\,166\,659, \\ &+ 4\,208\,420 \rightarrow 4\,208\,147, \end{aligned}$$

the sign giving the strand of D_k on which the occurrence lies and the arrow pointing to the operon it meets. The multiplicity-seven chains carry more k -mers than the chains of any other multiplicity over the operons: 4154 at $k = 150$, 5254 at $k = 100$, 6660 at $k = 51$ (check (M)).

Remark 7 (What was wrong with $5 + 2$, exactly). Nothing was wrong with it; it was the correct answer to a single-strand question. The content of Corollary 6 is that the question has a balanced-digraph answer too, and that the object which answers it is not the bidirected quotient but the double cover, on which the existing theorems already hold. The residual multiplicities 2, 3, 4, 5, 6 that still appear over the operons in D_k are the sub-families on which only some of the seven copies agree — the operons are not identical to one another, on either strand.

4 The involution dualises, it does not split

Proposition 8 (ρ and the opposite digraph). *ρ is an involution of the vertex set of D_k carrying the arc for a $(k+1)$ -mer x to the arc for $\bar{x}$ with its direction reversed. Hence ρ is an isomorphism $D_k \rightarrow D_k^{\text{op}}$, and it induces an isomorphism*

$$\mathcal{K}(D_k) \xrightarrow{\sim} \mathcal{K}(D_k)^\vee$$

onto the Pontryagin dual, i.e. a perfect pairing on $\mathcal{K}(D_k)$.

Proof. The prefix of $\bar{x}$ is the reverse complement of the suffix of x and vice versa, so the arc of $\bar{x}$ runs $\rho(\text{suffix}) \rightarrow \rho(\text{prefix})$: direction reverses, and ρ is an isomorphism onto the opposite digraph. For a balanced digraph $\Delta^{\text{op}} = D^+ - A^\top = \Delta^\top$, so $\mathcal{K}(D_k^{\text{op}}) = \text{Tor coker } \Delta_s^\top$ while $\mathcal{K}(D_k) = \text{Tor coker } \Delta_s$, and these two are canonically dual by the Bowen–Franks pairing [1, Lem. 1.5]. Composing gives the stated isomorphism. $\square$

Remark 9 (Against the eigenspace decomposition). [2, Open (c)] warned that $\mathcal{K}$ should not be split into ± 1 -eigenspaces of ρ , because that discards the 2-torsion. Proposition 8 is what happens instead: an involution that reverses arcs cannot act on $\mathcal{K}(D_k)$ at all — it does not preserve the presentation — but it identifies $\mathcal{K}(D_k)$ with its dual. No inversion of 2 occurs anywhere, and the 2-torsion is untouched. We do not determine whether the resulting pairing is symmetric or alternating; that question, and whether it is independent of the choice of sink, are left open.

k	graph	branch	chains	skeleton	$\mathcal{K}(\text{Sk})$	length-driven
12	G_k	2	1	1	0	$\mathbb{Z}/2$
	D_k	7	0	7	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	0
11	G_k	10	2	8	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	$(\mathbb{Z}/2)^2$
	D_k	30	7	23	$(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/49451$	$(\mathbb{Z}/2)^7$
10	G_k	47	8	37	$\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$	$(\mathbb{Z}/2)^{10}$
	D_k	125	23	97	$(\mathbb{Z}/4)^2 \oplus \mathbb{Z}/79624452132275189516228869$	$(\mathbb{Z}/2)^{28}$

Table 1: φX174 : the single-strand graph and its double cover. All entries exact (check (K)); the single-strand rows reproduce [4, Tab. 1].

5 Computations

Remark 10 (The phage’s single bit is single-strand). $\mathcal{K}(G_{12}) = \mathbb{Z}/2$: the 12-mer spectrum of φX174 leaves exactly two global reconstructions, and [3] located the bit as a pair of interleaved 12-mers. On the double-stranded spectrum the answer is different: $\mathcal{K}(D_{12}) = \mathbb{Z}/3 \oplus \mathbb{Z}/11 = \mathbb{Z}/33$, and it is entirely arrangement — D_{12} has no bundle chain at all, so the length-driven summand is trivial. The two statements do not conflict; they answer different questions, and the contrast is a fair measure of how much of the single-strand picture is an artefact of ignoring one strand.

k	branch	chains	skeleton	$\log_{10}$ length-driven	$\log_{10} \mathcal{K}(\text{Sk}) $	$\log_{10} \mathcal{K}(D_k) $
150	39 768	278	312	20 987.91	156.51	21 144.42
100	44 808	396	434	23 394.31	208.92	23 603.24
51	53 380	798	886	27 405.96	395.80	27 801.75

Table 2: *E. coli* K-12 MG1655, the double cover. Structural entries exact; $\log_{10} |\mathcal{K}(\text{Sk})|$ by float LU. Compare [6, Tab. 1]: on one strand the same k give skeletons of 115, 175 and 366 vertices and arrangement exponents 49.71, 71.11 and 143.17.

Two readings of Table 2. The skeleton roughly triples ($115 \rightarrow 312$ at $k = 150$) and the arrangement exponent does too ($49.71 \rightarrow 156.51$): inverted repeats are a large part of the irreducible ambiguity of this genome and the single-strand theory sees none of them. And $\log_{10} |\mathcal{K}(D_k)|$ exceeds twice the single-strand value by 166.6, 222.1 and 368.1 at $k = 150, 100, 51$ — a crude but honest measure of what double-strandedness adds beyond simply having two strands to reconstruct.

6 Verification

7 Scope

Proved. Lemma 2, Corollary 3, Proposition 4, Theorem 5, Corollary 6 and Proposition 8. All are short; the work is in noticing that the double cover is balanced and that this is enough.

What this is not. D_k is the *double cover*, not the bidirected quotient. The two are different objects and only the first is a digraph. In particular:

- $\mathcal{K}(D_k)$ does *not* count double-stranded reconstructions. By BEST, $|\mathcal{K}(D_k)| \prod_v (d^+(v) - 1)!$ is the number of arc-rooted Eulerian circuits of D_k , and such a circuit traverses all $2N$ arcs

key	check	mode
(B)	Lemma 2: D_k balanced, multiplicities add across strands	exact
(C)	Proposition 4: connected iff an inverted repeat of length $\geq k$, with witness	exact
(R)	Proposition 8: ρ is an involution with $A(\rho u, \rho v) = A(v, u)$	exact
(M)	Corollary 6: the multiplicity-seven chains and their seven operons	exact
(K)	Table 1: $\mathcal{K}(D_k)$ exactly, beside $\mathcal{K}(G_k)$	exact
(E)	Table 2	exact/numeric

Table 3: The verification battery (`rc_double_cover.py`), 6/6, about 33 seconds.

in one closed walk, which is not a genome. A double-stranded circular molecule corresponds instead to a decomposition of D_k into two ρ -conjugate circuits. Counting those is not solved here.

- No sandpile theory for the bidirected quotient is offered. The tools [8] named in [2, 3] remain the place to start, and the obstruction noted there — that reverse complementation reverses arc direction, so the quotient map is not a covering of digraphs and [7] does not apply — is untouched.
- The pairing of Proposition 8 is not identified as symmetric or alternating, and we have not checked that it is independent of the sink.

Numeric. $\log_{10} |\mathcal{K}(\text{Sk})|$ in Table 2 is a float LU determinant of a 311- to 885-dimensional matrix; everything else is exact integer work.

Open. (a) The two questions just listed. (b) Whether Theorem 5 has a converse: is every multiplicity- r family of D_k a strand-split family of S , and can a and b be recovered from D_k alone? (c) The 26-digit prime in Table 1 we do not interpret, as in [4].

Provenance

Unrefereed preprint. Every statement above is proved in full and every number is produced by `rc_double_cover.py` from the two public genomes. The task that prompted this note asked for a bidirected bundle-chain lemma yielding summands $(\mathbb{Z}/2r)^{s-1}$; no such lemma is needed for the double cover, and Theorem 5 shows the correct multiplicity is $a + b$, the number of copies on both strands together, not $2r$. The same task asked for “the algebraic proof that the seven operons participate in the skeleton as a single seven-copy family”; that is Corollary 6, with the caveat of Remark 7 — they do so in the double cover, and there are two such families exchanged by ρ , not one.

References

- [1] R. Chen, *Algorithmic non-commutative class field theory, VIII: the tower-level Hecke–Kirchberg limit and the non-commutative Iwasawa K -module*, 2026.
- [2] R. Chen, *Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs*, companion note (Bio 1, version 2), 2026.

- [3] R. Chen, *The sandpile group as a coordinate system on genome reconstructions*, companion note (Bio 2), 2026.
- [4] R. Chen, *The sandpile group of a de Bruijn graph splits along its repeats*, companion note (Bio 3), 2026.
- [5] R. Chen, *Multi-copy repeats: where the two-copy theory of assembly ambiguity stops*, companion note (Bio 4), 2026.
- [6] R. Chen, *The bundle-chain decomposition of a bacterial genome: Escherichia coli K-12 at five k-mer lengths*, companion note (Bio 5), 2026.
- [7] V. Reiner and D. Tseng, *Critical groups of covering, voltage and signed graphs*, Discrete Math. **318** (2014), 10–40.
- [8] T. Zaslavsky, *Signed graphs*, Discrete Appl. Math. **4** (1982), 47–74.